

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, TypeScript, JavaScript, SQL, C++, Java, Bash

LLM & Agentic Systems

LLM agents, tool use, function calling, prompt design, evaluation, automation tooling, provider routing/fallback, OpenAI / Groq APIs, Ollama, MCP

Full-Stack & Backend

React, Electron, Node.js, Express, FastAPI, REST APIs, async services, integration testing, MongoDB, PostgreSQL, Redis, SQLite

ML & NLP

PyTorch, HuggingFace Transformers, RAG, sentence-transformers, RRF, cross-encoder reranking, MMR, llama-index

Cloud & DevOps

AWS, GCP, Azure, Docker, Cloudflare Workers / Durable Objects / D1, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected and shipped a **full-stack internal tool** (React, Node.js, MongoDB) used daily by the AI team to annotate training data for production models; embedded with the team to learn their workflow, replaced spreadsheet workflows with secure role-based REST APIs and reviewer UIs, and **cut manual operations by 100+ hours/month while improving labeling throughput by 60%**.
- Acted as primary technical point of contact for the eTriage backend, collaborating across clinical, AI, and product stakeholders to scope, ship, and document features (auth, rate limiting, audit logging) used by **8+ UK clients** and processing **10,000+ monthly submissions**; contributed to **\$340K+** in revenue.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Shipped **LLM-powered context-aware replies** into a customer-facing chat product, iterating on prompt design and response-quality heuristics based on real usage; built **20+ Node.js + Fastify REST APIs** over PostgreSQL with input validation and clear contracts.

Projects

AgenticSearch - LLM-Powered End-to-End Automation Pipeline

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Designed and shipped an **AI-powered automation** that converts open-ended topic queries into structured entity tables end-to-end via typed retrieval planning, page rerank, deterministic parsers with LLM fallback, output verification, and **cell-level provenance** for every result.
- Hardened the service with **150+ automated tests**, provider routing/fallback across OpenAI and Groq, pipeline instrumentation, and a reviewer-facing trust UI for measuring extraction quality and iterating on prompt/heuristic changes.

cf-ai-research-scout - Full-Stack LLM Agent on Cloudflare Workers

github.com/Deva-1903/cf_ai_research_scout

TypeScript, Cloudflare Workers AI, AIChatAgent, Durable Objects, WorkflowEntrypoint, D1, WebSockets

- Built an **LLM agent** that fetches web sources, generates multi-step research digests, and streams responses to the client over WebSocket in real time, prototyped end-to-end with a TypeScript frontend on Cloudflare Workers and D1 storage.
- Designed durable workflow boundaries so multi-step agent tasks survive process restarts and tool failures; agent state and conversation history persist server-side without an external database.

AutoEval - Self-Improvement Agent with Reusable Eval Templates

github.com/Deva-1903/AutoEval

Python, IR Evaluation, Git Automation

- Built an **automation agent** that proposes new queries and labels, scores them against multiple reference systems, and accepts only updates that improve discriminative power on a holdout-heavy fitness objective, codifying what works as reusable templates and report cards committed back to Git.
- Added anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection) and Git-backed acceptance gates so every accepted change ships with an auditable report card, creating a **reusable playbook** other engineers can extend.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut). **Sonare** (Qualcomm Edge AI Hackathon 2025): offline sign ↔ speech cross-platform app shipped under hackathon time pressure.
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.